

ROUTER AND THE NETWORK LAYER

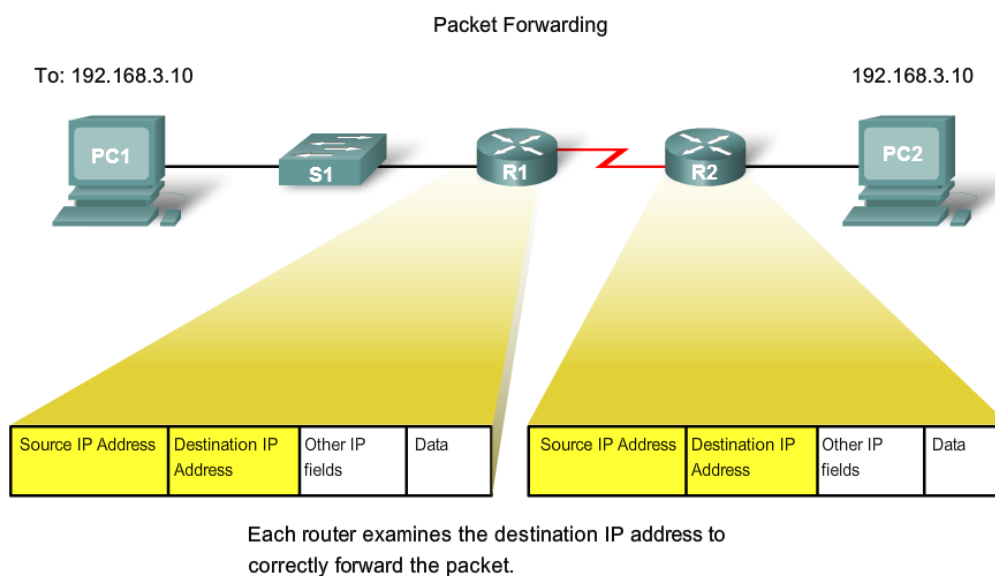
Routers and the Network Layer

The main purpose of a router is to connect multiple networks and forward packets destined either for its own networks or other networks. A router is considered a Layer 3 device because its primary forwarding decision is based on the information in the Layer 3 IP packet, specifically the destination IP address. This process is known as routing.

When a router receives a packet, it examines its destination IP address. If the destination IP address does not belong to any of the router's directly connected networks, the router must forward this packet to another router. In the figure, R1 examines the destination IP address of the packet. After searching the routing table, R1 forwards the packet onto R2. When R2 receives the packet, it also examines the packet's destination IP address. After searching its routing table, R2 forwards the packet out its directly connected Ethernet network to PC2.

When each router receives a packet, it searches its routing table to find the best match between the destination IP address of the packet and one of the network addresses in the routing table. Once a match is found, the packet is encapsulated in the layer 2 data link frame for that outgoing interface. The type of data link encapsulation depends on the type of interface, such as Ethernet or HDLC.

Eventually the packet reaches a router that is part of a network that matches the destination IP address of the packet. In this example, router R2 receives the packet from R1. R2 forwards the packet out its Ethernet interface, which belongs to the same network as the destination device, PC2.



Routers Operate at Layers 1, 2, and 3

A router makes its primary forwarding decision at Layer 3, but also participates in Layer 1 and Layer 2 processes as well. After a router has examined the destination IP address of a packet and consulted its routing table to make its forwarding decision, it can forward that packet out the appropriate interface toward its destination. The router encapsulates the Layer 3 IP packet into the data portion of a Layer 2 data link frame appropriate for the exit interface. The type of frame can be an Ethernet, HDLC, or some other Layer 2 encapsulation - whatever encapsulation is used on that particular interface. The Layer 2 frame is encoded into the Layer 1 physical signals that are used to represent bits over the physical link.

In the figure in page 3, PC1 operates at all seven layers, encapsulating the data and sending the frame out as a stream of encoded bits to R1, its default gateway.

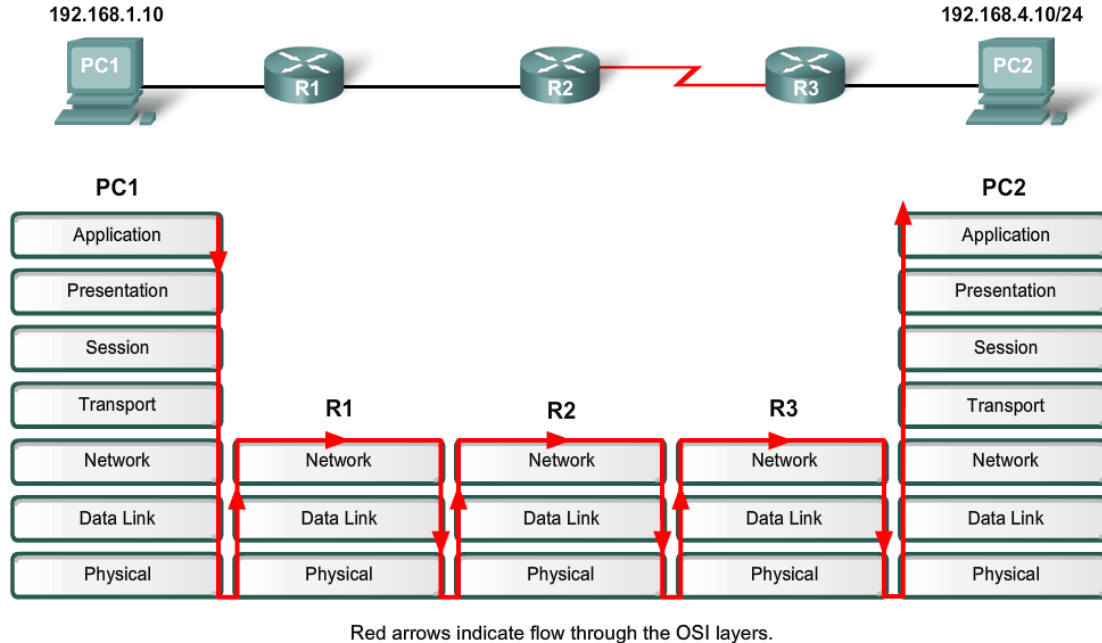
R1 receives the stream of encoded bits on its interface. The bits are decoded and passed up to Layer 2, where R1 decapsulates the frame. The router examines the destination address of the data link frame to determine if it matches the receiving interface, including a broadcast or multicast address. If there is a match with the data portion of the frame, the IP packet is passed up to Layer 3, where R1 makes its routing decision. R1 then re-encapsulates the packet into a new Layer 2 data link frame and forwards it out the outbound interface as a stream of encoded bits.

R2 receives the stream of bits, and the process repeats itself. R2 decapsulates the frame and passes the data portion of the frame, the IP packet, to Layer 3 where R2 makes its routing decision. R2 then re-encapsulates the packet into a new Layer 2 data link frame and forwards it out the outbound interface as a stream of encoded bits.

This process is repeated once again by router R3, which forwards the IP packet, encapsulated inside a data link frame and encoded as bits, to PC2.

Each router in the path from source to destination performs this same process of decapsulation, searching the routing table, and then re-encapsulation.

Router Operates at Layers 1, 2, and 3



BUILDING THE ROUTING TABLE

Introducing the Routing Table

The primary function of a router is to forward a packet toward its destination network, which is the destination IP address of the packet. To do this, a router needs to search the routing information stored in its routing table.

A routing table is a data file in RAM that is used to store route information about directly connected and remote networks. The routing table contains network/next hop associations. These associations tell a router that a particular destination can be optimally reached by sending the packet to a specific router that represents the "next hop" on the way to the final destination. The next hop association can also be the outgoing or exit interface to the final destination.

The network/exit-interface association can also represent the destination network address of the IP packet. This association occurs on the router's directly connected networks.

A directly connected network is a network that is directly attached to one of the router interfaces. When a router interface is configured with an IP address and subnet mask, the interface becomes a host on that attached network. The network address and subnet mask of the interface, along with the interface type and number, are entered into the routing table as a directly connected network. When a router forwards a packet to a host, such as a web server, that host is on the same network as a router's directly connected network.

A remote network is a network that is not directly connected to the router. In other words, a remote network is a network that can only be reached by sending the packet to another router. Remote networks are added to the routing table using either a dynamic routing protocol or by configuring static routes. Dynamic routes are routes to remote networks that were learned automatically by the router, using a dynamic routing protocol. Static routes are routes to networks that a network administrator manually configured.

Directly Connected Networks

Adding a Connected Network to the Routing Table

Before any static or dynamic routing is configured on a router, the router only knows about its own directly connected networks. These are the only networks that are displayed in the routing table until static or dynamic routing is configured. Directly connected networks are of prime importance for routing decisions. Static and dynamic routes cannot exist in the routing table without a router's own directly connected networks.

Static Routing

Remote networks are added to the routing table either by configuring static routes or enabling a dynamic routing protocol. When the IOS learns about a remote network and the interface that it will use to reach that network, it adds that route to the routing table as long as the exit interface is enabled.

A static route includes the network address and subnet mask of the remote network, along with the IP address of the next-hop router or exit interface.

Typically, most routing tables contain a combination of static routes and dynamic routes. But, as stated earlier, the routing table must first contain the directly connected networks used to access these remote networks before any static or dynamic routing can be used.

Dynamic Routing

Remote networks can also be added to the routing table by using a dynamic routing protocol. In the figure, R1 has automatically learned about the 192.168.4.0/24 network from R2 through the dynamic routing protocol, RIP (Routing Information Protocol).

Dynamic routing protocols are used by routers to share information about the reachability and status of remote networks. Dynamic routing protocols perform several activities, including:

- Network discovery
- Updating and maintaining routing tables

Automatic Network Discovery

Network discovery is the ability of a routing protocol to share information about the networks that it knows about with other routers that are also using the same routing protocol. Instead of configuring static routes to remote networks on every router, a dynamic routing protocol allows the routers to automatically learn about these networks from other routers. These networks - and the best path to each network - are added to the router's routing table and denoted as a network learned by a specific dynamic routing protocol.

Maintaining Routing Tables

After the initial network discovery, dynamic routing protocols update and maintain the networks in their routing tables. Dynamic routing protocols not only make a best path determination to various networks, they will also determine a new best path if the initial path becomes unusable (or if the topology changes). For these reasons, dynamic routing protocols have an advantage over static routes. Routers that use dynamic routing protocols automatically share routing information with other routers and compensate for any topology changes without involving the network administrator.

IP Routing Protocols

There are several dynamic routing protocols for IP. Here are some of the more common dynamic routing protocols for routing IP packets:

- RIP (Routing Information Protocol)
- IGRP (Interior Gateway Routing Protocol)
- EIGRP (Enhanced Interior Gateway Routing Protocol)
- OSPF (Open Shortest Path First)
- IS-IS (Intermediate System-to-Intermediate System)
- BGP (Border Gateway Protocol)

PATH DETERMINATION AND SWITCHING FUNCTIONS

Best Path And Metric

The best path is selected by a routing protocol based on the value or metric it uses to determine the distance to reach a network. Some routing protocols, such as RIP, use simple hop-count, which the number of routers between a router and the destination network. Other routing protocols, such as OSPF, determine the shortest path by examining the bandwidth of the links, and using the links with the fastest bandwidth from a router to the destination network.

Dynamic routing protocols typically use their own rules and metrics to build and update routing tables. A metric is the quantitative value used to measure the distance to a given route. The best

path to a network is the path with the lowest metric. For example, a router will prefer a path that is 5 hops away over a path that is 10 hops away.

The primary objective of the routing protocol is to determine the best paths for each route to include in the routing table. The routing algorithm generates a value, or a metric, for each path through the network. Metrics can be based on either a single characteristic or several characteristics of a path. Some routing protocols can base route selection on multiple metrics, combining them into a single metric. The smaller the value of the metric, the better the path.

Path Determination

Packet forwarding involves two functions:

- Path determination function
- Switching function

The path determination function is the process of how the router determines which path to use when forwarding a packet. To determine the best path, the router searches its routing table for a network address that matches the packet's destination IP address.

Switching Function

After the router has determined the exit interface using the path determination function, the router needs to encapsulate the packet into the data link frame of the outgoing interface.

The switching function is the process used by a router to accept a packet on one interface and forward it out another interface. A key responsibility of the switching function is to encapsulate packets in the appropriate data link frame type for the outgoing data link.

What does a router do with a packet received from one network and destined for another network? The router performs the following three major steps:

1. Decapsulates the Layer 3 packet by removing the Layer 2 frame header and trailer.
2. Examines the destination IP address of the IP packet to find the best path in the routing table.
3. Encapsulates Layer 3 packet into a new Layer 2 frame and forwards the frame out the exit interface.

As the IP packet is decapsulated from one Layer 2 frame and encapsulated into a new Layer 2 frame, the data link destination address and source address will change as the packet is forwarded from one router to the next. The Layer 2 data link source address represents the Layer 2 address of the outbound interface. The Layer 2 destination address represents the Layer 2 address of the

next-hop router. If the next hop is the final destination device, it will be the Layer 2 address of that device.

It is very likely that the packet will be encapsulated in a different type of Layer 2 frame than the one in which it was received. For example, the packet might be received by the router on a FastEthernet interface, encapsulated in an Ethernet frame, and forwarded out a serial interface encapsulated in a PPP frame.

Remember, as a packet travels from the source device to the final destination device, the Layer 3 IP addresses do not change. However, the Layer 2 data link addresses change at every hop as the packet is decapsulated and re-encapsulated in a new frame by each router.

Path Determination and Switching Function Details

Step 1:

PC1 has a packet to be sent to PC2

PC1 encapsulates the IP packet into an Ethernet frame with the destination MAC address of R1's FastEthernet 0/0 interface.

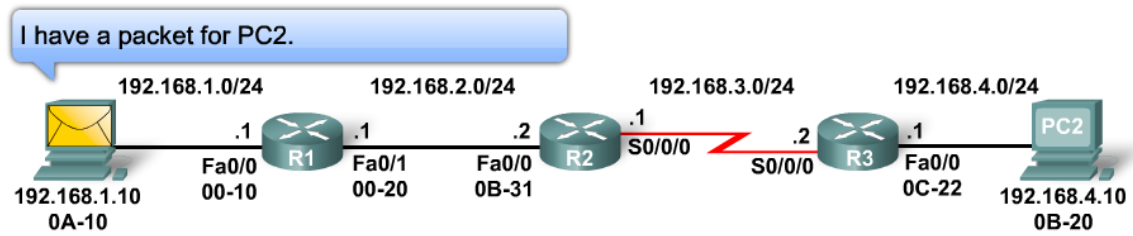
How does PC1 know to forward to packet to R1 and not directly to PC2? PC1 has determined that the IP source and IP destination addresses are on different networks.

PC1 knows the network it belongs to by doing an AND operation on its own IP address and subnet mask, which results in its network address. PC1 does this same AND operation using the packet destination IP address and the PC1 subnet mask. If the result is the same as its own network, PC1 knows that the destination IP address is on its own network and it does not need to forward the packet to the default gateway, the router. If the AND operation results in a different network address, PC1 knows that the destination IP address is not on its own network and that it must forward this packet to the default gateway, the router.

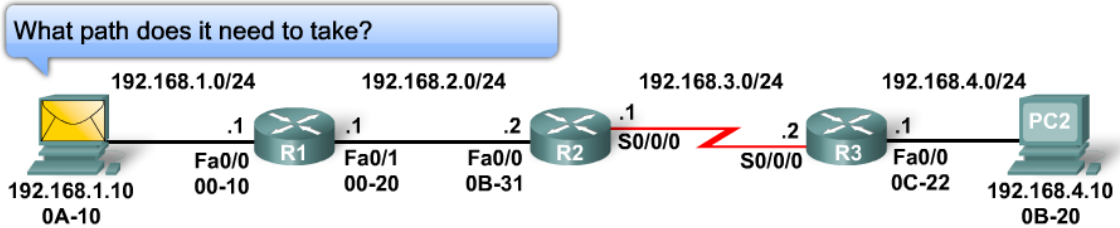
How does PC1 determine the MAC address of the default gateway, router R1? PC1 checks its ARP table for the IP address of the default gateway and its associated MAC address.

What if this entry does not exist in the ARP table? PC1 sends an ARP request and router R1 sends back an ARP reply.

A day in the life of a packet: Step 1



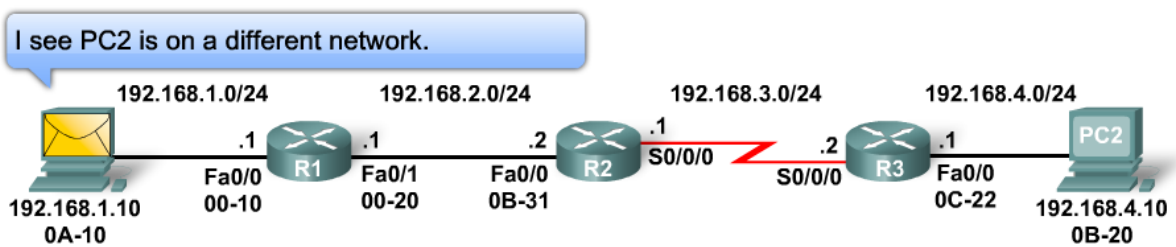
A day in the life of a packet: Step 1



Packet's Layer 3 data

Source IP	Dest. IP	IP fields	Data
192.168.1.10	192.168.4.10		

A day in the life of a packet: Step 1

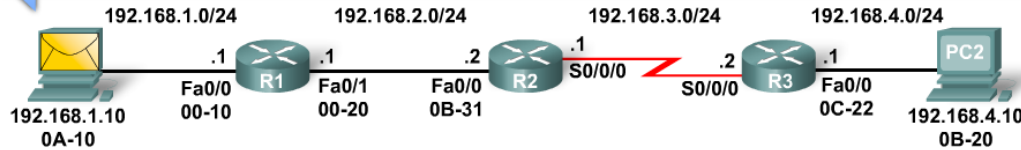


Packet's Layer 3 data

Source IP	Dest. IP	IP fields	Data
192.168.1.10	192.168.4.10		

A day in the life of a packet: Step 1

Since PC2 is on different network, I'll encapsulate packet and send it to the router on MY network. Let me find that MAC address....



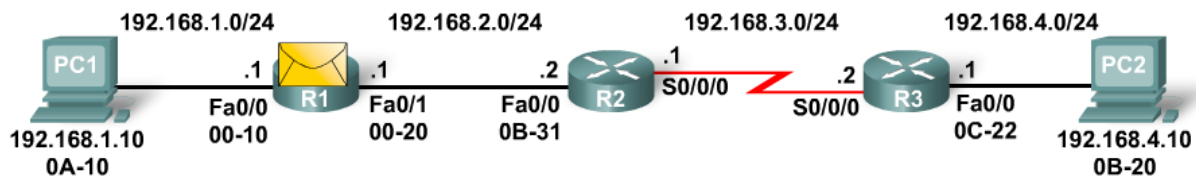
Layer 2 Data Link Frame

Dest. MAC 00-10	Source MAC 0A-10	Type 800	Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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Packet's Layer 3 data

PC1's ARP Cache for R1	
IP Address	MAC Address
192.168.1.1	00-10

A day in the life of a packet: Step 1



Step 2: Router R1 receives the Ethernet frame

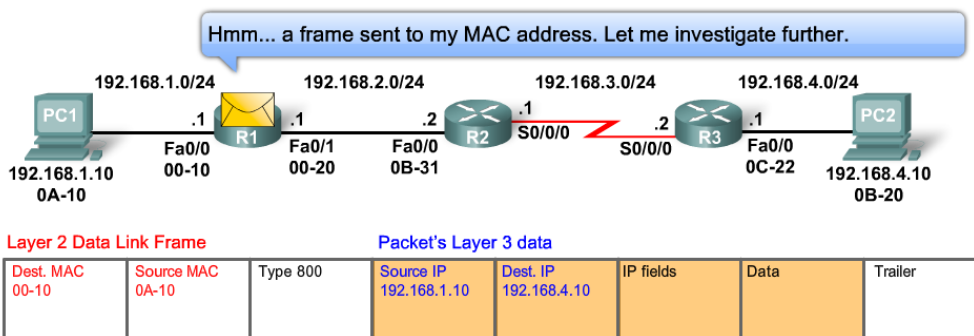
1. Router R1 examines the destination MAC address, which matches the MAC address of the receiving interface, FastEthernet 0/0. R1 will therefore copy the frame into its buffer.
2. R1 sees that the Ethernet Type field is 0x800, which means that the Ethernet frame contains an IP packet in the data portion of the frame.
3. R1 decapsulates the Ethernet frame.
4. Because the destination IP address of the packet does not match any of R1's directly connected networks, the router consults its routing table to route this packet. R1 searches the routing table for a network address and subnet mask that would include this packet's destination IP address as a host address on that network. In this example, the routing table has a route for the 192.168.4.0/24 network. The destination IP address of the packet is 192.168.4.10, which is a host IP address on that network.

R1's route to the 192.168.4.0/24 network has a next-hop IP address of 192.168.2.2 and an exit interface of FastEthernet 0/1. This means that the IP packet will be encapsulated in a new Ethernet frame with the destination MAC address of the next-hop router's IP address. Because the exit interface is on an Ethernet network, R1 must resolve the next-hop IP address with a destination MAC address.

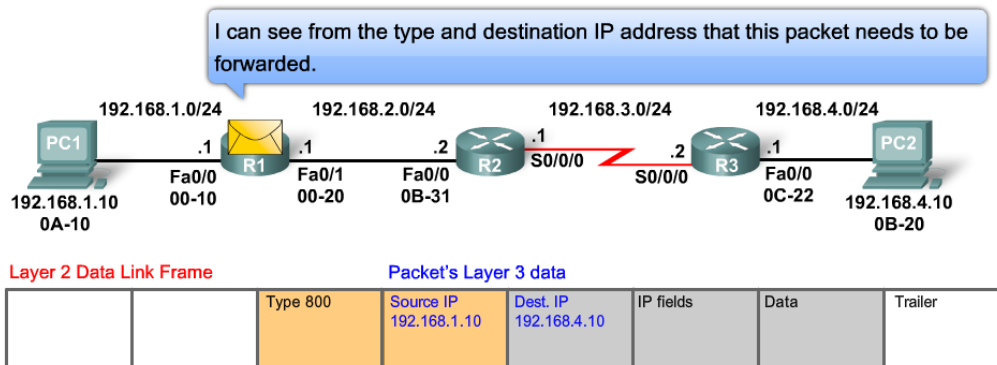
5. R1 looks up the next-hop IP address of 192.168.2.2 in its ARP cache for its FastEthernet 0/1 interface. If the entry is not in the ARP cache, R1 sends an ARP request out its FastEthernet 0/1 interface. R2 sends back an ARP reply. R1 then updates its ARP cache with an entry for 192.168.2.2 and the associated MAC address.

6. The IP packet is now encapsulated into a new Ethernet frame and forwarded out R1's FastEthernet 0/1 interface.

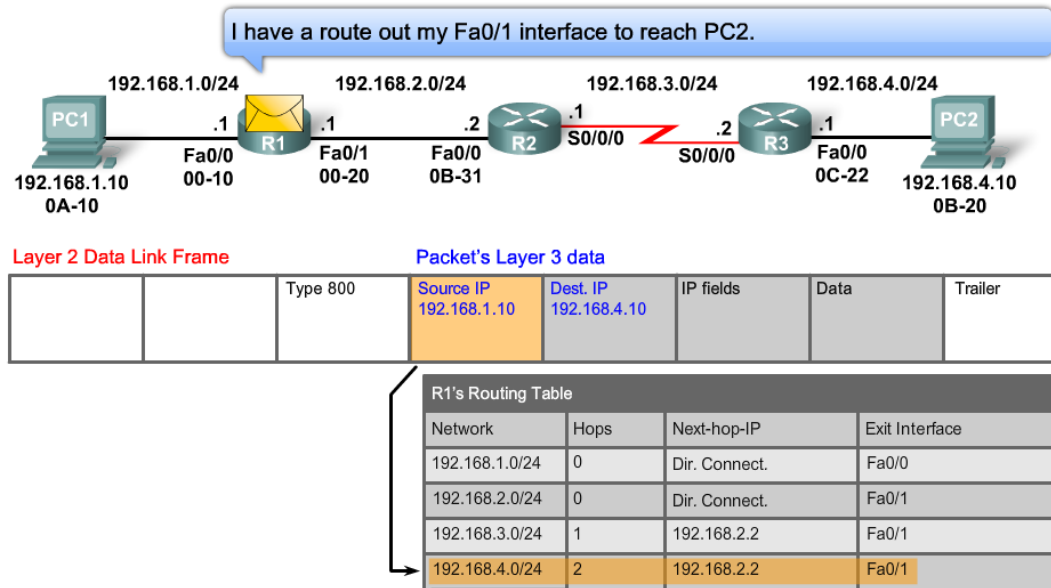
A day in the life of a packet: Step 2



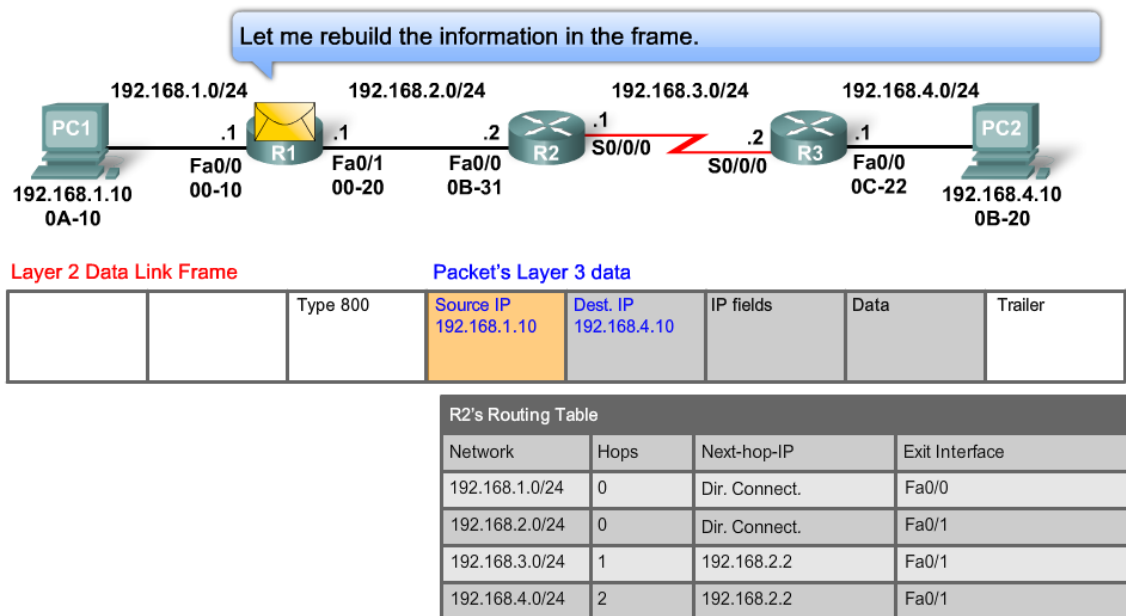
A day in the life of a packet: Step 2



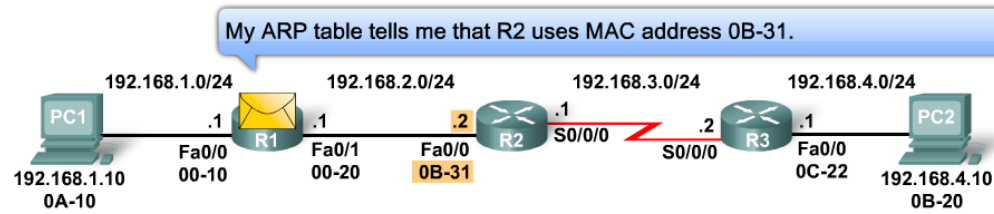
A day in the life of a packet: Step 2



A day in the life of a packet: Step 2



A day in the life of a packet: Step 2



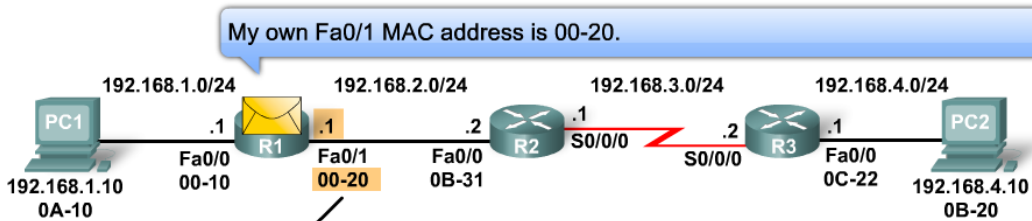
Layer 2 Data Link Frame

Dest. MAC	Type	Source IP	Dest. IP	IP fields	Data	Trailer
0B-31	800	192.168.1.10	192.168.4.10			

Packet's Layer 3 data

R1's ARP Cache		R2's Routing Table			
IP Address	MAC Address	Network	Hops	Next-hop-IP	Exit Interface
192.168.2.2	0B-31	192.168.1.0/24	0	Dir. Connect.	Fa0/0
		192.168.2.0/24	0	Dir. Connect.	Fa0/1
		192.168.3.0/24	1	192.168.2.2	Fa0/1
		192.168.4.0/24	2	192.168.2.2	Fa0/1

A day in the life of a packet: Step 2



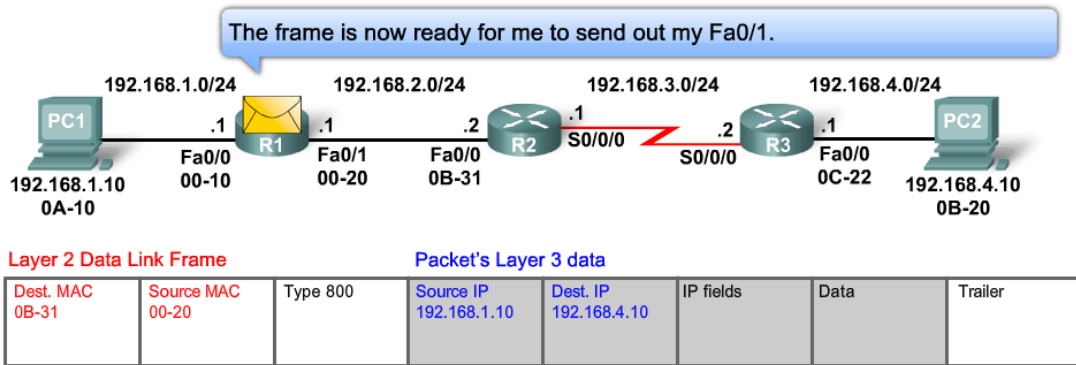
Layer 2 Data Link Frame

Dest. MAC	Source MAC	Type	Source IP	Dest. IP	IP fields	Data	Trailer
0B-31	00-20	800	192.168.1.10	192.168.4.10			

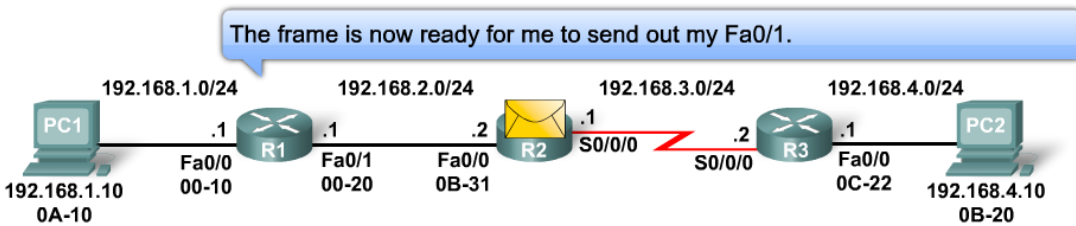
Packet's Layer 3 data

R2's Routing Table			
Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	0	Dir. Connect.	Fa0/0
192.168.2.0/24	0	Dir. Connect.	Fa0/1
192.168.3.0/24	1	192.168.2.2	Fa0/1
192.168.4.0/24	2	192.168.2.2	Fa0/1

A day in the life of a packet: Step 2

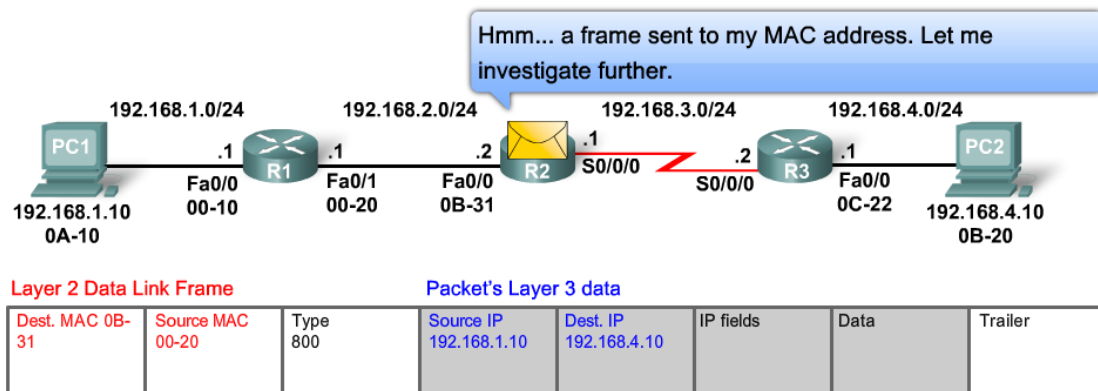


A day in the life of a packet: Step 2

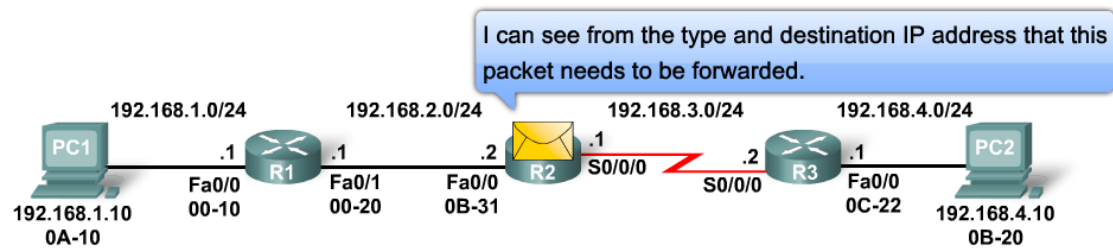


Step 3: Packet arrives at router R2

A day in the life of a packet: Step 3



A day in the life of a packet: Step 3

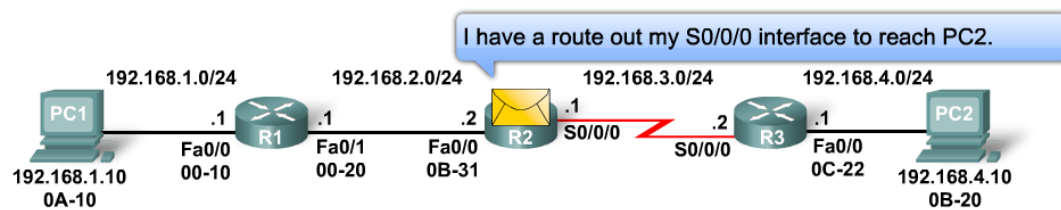


Layer 2 Data Link Frame

Packet's Layer 3 data

		Type 800	Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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A day in the life of a packet: Step 3



Layer 2 Data Link Frame

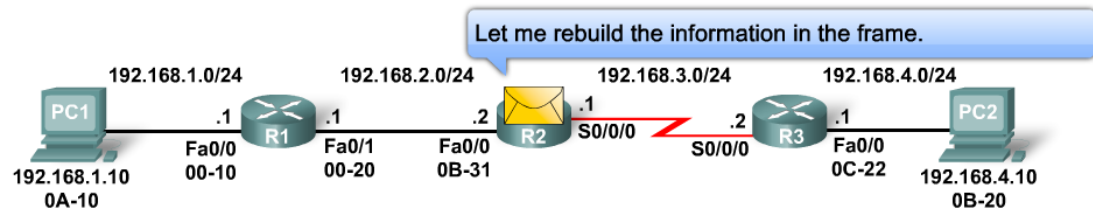
Packet's Layer 3 data

			Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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R2's Routing Table

Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	1	192.168.2.1	Fa0/0
192.168.2.0/24	0	Dir. Connect.	Fa0/0
192.168.3.0/24	0	Dir. Connect.	S0/0/0
192.168.4.0/24	1	192.168.3.2	S0/0/0

A day in the life of a packet: Step 3



Layer 2 Data Link Frame

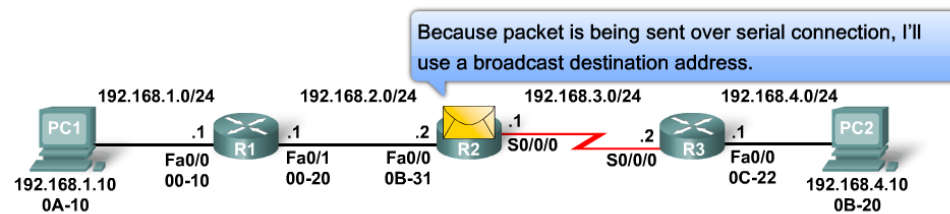
			Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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Packet's Layer 3 data

R2's Routing Table

Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	1	192.168.2.1	Fa0/0
192.168.2.0/24	0	Dir. Connect.	Fa0/0
192.168.3.0/24	0	Dir. Connect.	S0/0/0
192.168.4.0/24	1	192.168.3.2	S0/0/0

A day in the life of a packet: Step 3



Layer 2 Data Link Frame

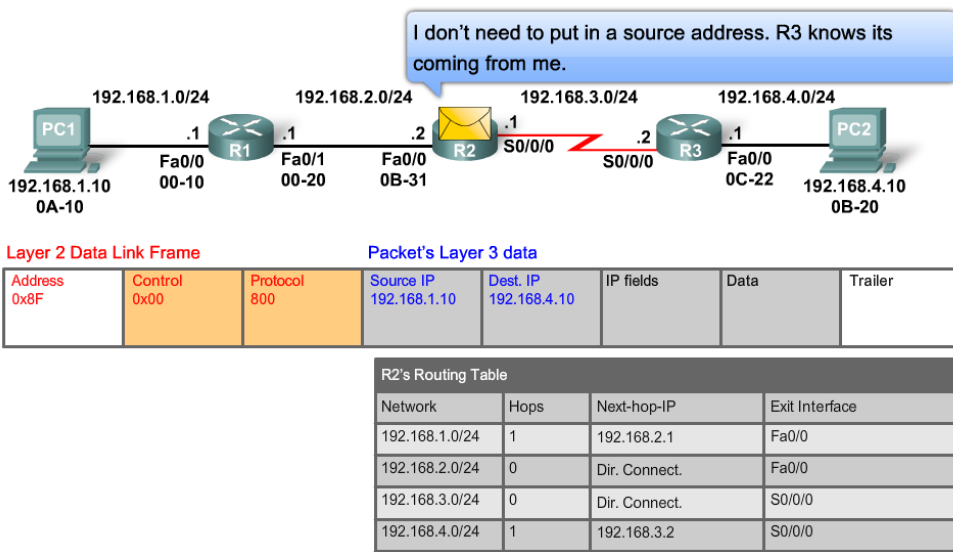
Address 0x8F			Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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Packet's Layer 3 data

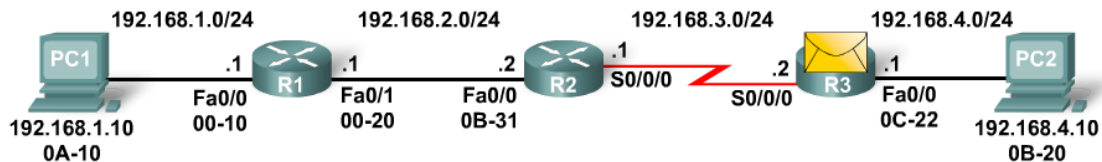
R2's Routing Table

Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	1	192.168.2.1	Fa0/0
192.168.2.0/24	0	Dir. Connect.	Fa0/0
192.168.3.0/24	0	Dir. Connect.	S0/0/0
192.168.4.0/24	1	192.168.3.2	S0/0/0

A day in the life of a packet: Step 3



A day in the life of a packet: Step 3



Step 4: The packet arrives at R3

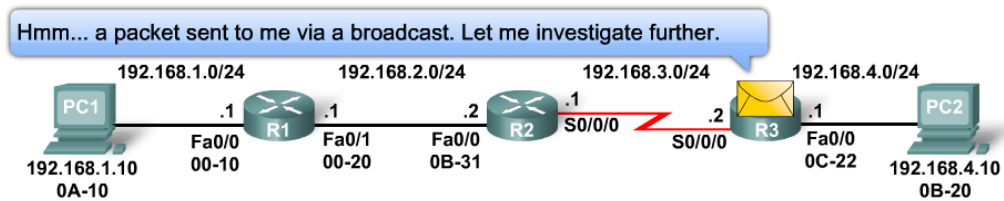
1. R3 receives and copies the data link PPP frame into its buffer.
2. R3 decapsulates the data link PPP frame.
3. R3 searches the routing table for the destination IP address of the packet. The search of the routing table results in a network that is one of R3's directly connected networks. This means that the packet can be sent directly to the destination device and does not need to be sent to another router.

Because the exit interface is a directly connected Ethernet network, R3 needs to resolve the destination IP address of the packet with a destination MAC address.

4. R3 searches for the packet's destination IP address of 192.168.4.10 in its ARP cache. If the entry is not in the ARP cache, R3 sends an ARP request out its FastEthernet 0/0 interface. PC2 sends back an ARP reply with its MAC address. R3 updates its ARP cache with an entry for 192.168.4.10 and the MAC address that was returned in the ARP reply.

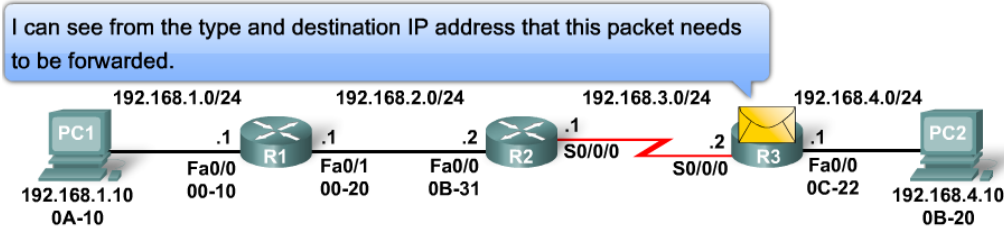
5. The IP packet is encapsulated into a new data link, Ethernet frame and sent out R3's FastEthernet 0/0 interface.

A day in the life of a packet: Step 4



Layer 2 Data Link Frame			Packet's Layer 3 data				
Address 0x8F	Control 0x00	Protocol 800	Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer

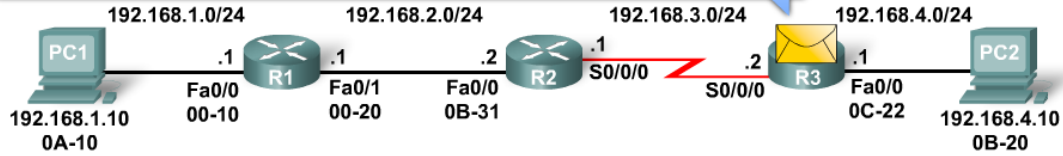
A day in the life of a packet: Step 4



Layer 2 Data Link Frame			Packet's Layer 3 data				
			Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer

A day in the life of a packet: Step 4

I have a route out my Fa0/0 interface to reach PC2.



Layer 2 Data Link Frame

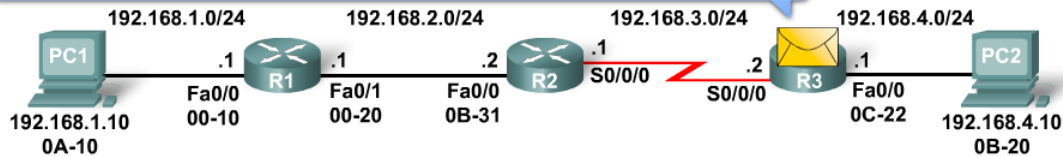
Packet's Layer 3 data

		Type 800	Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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R3's Routing Table			
Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	2	192.168.3.1	S0/0/0
192.168.2.0/24	1	192.162.3.1	S0/0/0
192.168.3.0/24	0	Dir. Connect.	S0/0/0
192.168.4.0/24	0	Dir. Connect.	Fa0/0

A day in the life of a packet: Step 4

Let me rebuild the information in the frame.



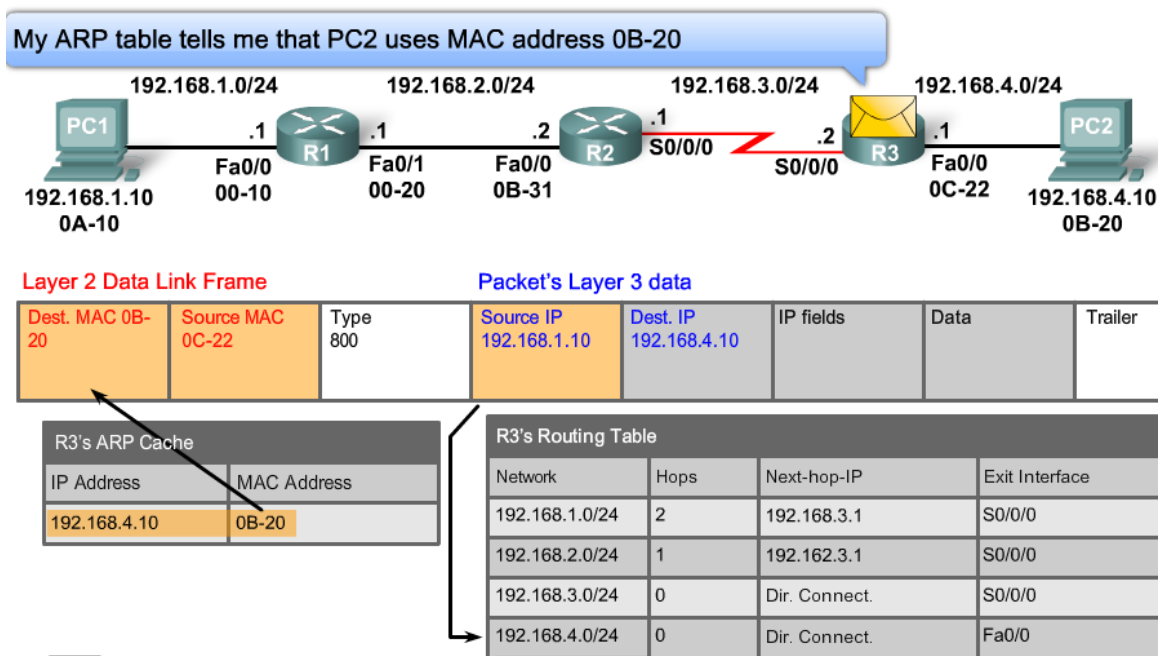
Layer 2 Data Link Frame

Packet's Layer 3 data

Dest. MAC	Source MAC	Type 800	Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	
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R3's Routing Table			
Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	2	192.168.3.1	S0/0/0
192.168.2.0/24	1	192.162.3.1	S0/0/0
192.168.3.0/24	0	Dir. Connect.	S0/0/0
192.168.4.0/24	0	Dir. Connect.	Fa0/0

A day in the life of a packet: Step 4



Step 5: The Ethernet Frame with encapsulated IP packet arrives at PC2

1. PC2 examines the destination MAC address, which matches the MAC address of the receiving interface, its Ethernet NIC. PC2 will therefore copy the rest of the frame into its buffer.
2. PC2 sees that the Ethernet Type field is 0x800, which means that the Ethernet frame contains an IP packet in the data portion of the frame.
3. PC2 decapsulates the Ethernet frame and passes the IP packet to the IP process of its operating system.

A day in the life of a packet: Step 4

